

A RESEARCH MONOGRAPH OF THE SERIES ON INTEGRATED AUTONOMOUS INTELLIGENCE

FROM TAX CODES TO GOVERNED STRUCTURAL INTELLIGENCE

The Tax Planning Exo-Brain as a Pedagogical Architecture
for Multi-Jurisdictional Institutional Cognition

MEMORY · GOVERNANCE · CONTROLLED AUTONOMY

*“Computation may enlarge the space of possible interpretations, but
only governance may enlarge the space of permitted action.”*

ALEJANDRO REYNOSO

RESEARCH MONOGRAPH · JULY 2026

Contents

ABSTRACT	1
1. Introduction: Tax Intelligence Is Not Rate Shopping	2
2. The Experimental Setting: Two Universes, One Portfolio, and a Durable Memory	3
3. Conceptual Foundations	5
4. The Eleven-Stage Maturity Ladder	7
5. A Formal Model of the Exo-Brain	12
6. What the Laboratory Teaches About Autonomous Systems	15
7. The Architecture of Authority	16
8. Why the Order of the Baby Steps Matters	18
9. Evaluation: What the Prototype Demonstrates	19
10. Limitations and Research Boundaries	20
11. Productionization Roadmap	20
12. Conclusion: The Objective Is Governed Adaptation	22
Appendix A. The Eleven Baby Steps at a Glance	22
Appendix B. Source Materials	23

ABSTRACT

This paper develops a complete pedagogical interpretation of an eleven-stage laboratory for constructing a governed tax-planning exo-brain. The laboratory begins with a synthetic external universe of 100 tax-code modules across 20 fictional jurisdictions and an internal universe of 10 complex conglomerates comprising 154 legal entities, 144 ownership relationships, and 226 intercompany transactions. It ends with a read-only operating application that integrates rule versioning, enterprise topology, recommendation lineage, stress testing, evidence diligence, constrained optimization, adviser-process design, controlled instruction dry runs, quarterly change detection, and human review queues. At no stage does the prototype authorize a filing, transaction, restructuring, communication, adviser appointment, or implementation.

The central argument is that advanced tax intelligence should not be understood as a machine that searches for the lowest statutory rate. Multi-jurisdictional tax planning is a governed state-transition problem in which legal rules, enterprise facts, strategic objectives, economic outcomes, evidentiary quality, operating substance, institutional decisions, and permissions must remain distinct but connected. A proposed structure has no institutional meaning unless the system can identify the rule versions on which it relies, the enterprise facts it assumes, the claims connecting those facts to the proposal, the contradictions that constrain reliance, the stress scenarios under which the structure becomes fragile, the human decision that authorizes the next analytical operation, and the actions that remain prohibited.

The eleven Baby Steps form a maturity ladder. Step 0 creates memory without judgment. Step 1 completes the first governed design loop. Step 2 tests generalization and fragility. Step 3 freezes provenance and converts narrative reasoning into atomic claims and contradictions. Step 4 reconstructs the governed state as a tax-committee product. Step 5 turns diligence into a causal evidence-refresh process. Step 6 performs multi-objective optimization inside legal, economic, substance, documentation, and complexity constraints. Step 7 designs a conflict- and independence-aware adviser process. Step 8 simulates instruction and implementation while proving non-action. Step 9 introduces quarterly legal change and universe growth without automatic revisionism. Step 10 exposes the accumulated intelligence through a read-only application boundary.

The paper reconstructs this architecture, formalizes its principal mechanisms, and explains why its sequence matters. It distinguishes legal information from enterprise facts, analytical preference from institutional position, confidence from permission, optimization from authorization, learning from historical erasure, and simulation from execution. The result is a general architecture for semi-autonomous systems in high-stakes institutional settings: computation may enlarge the space of possible interpretations, but only governance may enlarge the space of permitted action.

Keywords: exo-brain; tax planning; autonomous systems; multi-jurisdictional taxation; institutional cognition; knowledge architecture; provenance; substance; constrained optimization; contradiction control; human-in-the-loop governance; versioned legal intelligence.

1. Introduction: Tax Intelligence Is Not Rate Shopping

The most naive representation of international tax planning is an optimization problem over statutory rates. An enterprise has income, jurisdictions impose different taxes, and the planner seeks a structure that minimizes the resulting burden. This representation is attractive because it is computationally clean. It is also institutionally inadequate.

A real tax structure exists at the intersection of several systems. Law defines corporate income tax, withholding, transfer-pricing requirements, controlled-foreign-company rules, minimum-tax obligations, anti-avoidance standards, treaty positions, filing duties, and documentation requirements. The enterprise contributes ownership, people, functions, assets, risks, decision rights, contracts, financing flows, intellectual property, commercial objectives, and implementation capacity. Evidence determines which of those facts can be defended. Governance determines who may accept a position, commission more work, instruct an adviser, communicate externally, file a return, or implement a restructuring. Time changes all of them.

The resulting analytical problem is not merely: "Which structure produces the lowest projected tax?" It is:

- ▶ Which current rule versions apply to which entities, transactions, and functions?
- ▶ Which enterprise facts are observed, inferred, assumed, or still missing?
- ▶ Which candidate structure fits the business model rather than merely the tax model?
- ▶ Which legal and operational constraints are compensatory, and which are hard prohibitions?
- ▶ How does the proposal perform when rates, withholding, substance costs, or strategic objectives change?
- ▶ Which claims support the recommendation, and what contradictory claims constrain it?
- ▶ Which human decision permits the next analytical step?
- ▶ What must remain impossible until a separate authority changes the permission state?
- ▶ When the law changes, which parts of the portfolio require review and which can remain untouched?

These are questions of architecture before they are questions of calculation.

The Tax Planning Exo-Brain laboratory was designed to make this architecture visible. It does not claim to provide real tax advice. Every jurisdiction, rule, rate, company, adviser, recipient, transaction, and outcome is synthetic. The artificiality is deliberate. It creates a controlled environment in which one can study how a tax-intelligence system should remember, reason, challenge, decide, update, and remain bounded without confusing pedagogical machinery with professional reliance.

The term **exo-brain** is used precisely. The system is not an autonomous tax professional and not an oracle that emits conclusions from a prompt. It is a durable cognitive extension of an institution. It externalizes a portion of the institution's memory into governed objects: jurisdictions, tax codes, conglomerates, entities, transactions, sources, claims, contradictions, recommendations, scenarios, decisions, permissions, advisers, recipients, review queues, and historical versions. The system becomes valuable not because it produces one impressive

answer, but because it preserves the conditions under which answers may be inspected, challenged, revised, or held.

The project advances through eleven notebooks, numbered Baby Step 0 through Baby Step 10. Each notebook receives an accepted prior state, performs one bounded experiment, writes a new version of the vault, validates counts and invariants, records a decision boundary, and preserves the earlier state. This cumulative discipline produces three transformations.

1. **From scattered information to governed memory.** Tax rules and enterprise facts become separate but linked universes with stable identifiers, explicit relationships, version history, and deterministic validation.
2. **From governed memory to defensible institutional analysis.** Candidate structures become versioned recommendations, stress tests, atomic claims, contradiction registers, committee products, diligence records, and constrained design artifacts.
3. **From analysis to a bounded operating architecture.** Adviser selection, disclosure, dry-run instruction, legal change, review queues, playbooks, health checks, and a read-only application are integrated without silently creating execution authority.

The progression is pedagogical because each stage corrects a characteristic error. Step 0 prevents conclusions from being smuggled into the baseline. Step 1 prevents retrieval scores from masquerading as advice. Step 2 prevents a globally uniform model from concealing local fragility. Step 3 prevents polished narratives from outrunning their evidence. Step 4 prevents presentation from diverging from governed state. Step 5 prevents diligence from becoming document accumulation. Step 6 prevents economic optimization from overriding legal and operational feasibility. Step 7 prevents a high-ranked provider from bypassing conflict and independence gates. Step 8 prevents workflow simulation from becoming accidental action. Step 9 prevents continuous updating from becoming automatic revisionism. Step 10 prevents usability from creating an uncontrolled write path.

The paper explains the cumulative logic of these stages and derives a more general proposition: an advanced institutional AI system should be judged not only by the quality of its outputs, but also by the integrity of its memory, the visibility of its uncertainty, the separability of its authority layers, and its capacity to remain inactive when action has not been authorized.

2. The Experimental Setting: Two Universes, One Portfolio, and a Durable Memory

The laboratory begins with two universes that must be connected without being conflated.

The **external rule universe** contains 20 fictional jurisdictions and 100 synthetic tax-code modules, five per jurisdiction. The module taxonomy covers corporate income tax, withholding tax, transfer pricing, controlled-foreign-company rules, and global minimum tax. Each module is a versioned legal object with a stable identifier, jurisdiction, category, rate or threshold, effective period, status, and relationships to other legal objects.

The **internal enterprise universe** initially contains 10 synthetic conglomerates. Together they contain 154 legal entities, 144 ownership edges, and 226 intercompany transactions. The conglomerates represent heterogeneous business models: advanced manufacturing, digital

services, life sciences, consumer products, infrastructure, energy, financial technology, logistics, media and data, and industrial software. Their entity graphs encode ownership and control; their transaction graphs encode commercial and financial flows; their narrative records describe functions, assets, risks, strategy, constraints, and governance.

This separation is more than an organizational convenience. A tax rule is not a fact about a company. A company fact is not a legal conclusion. The proposition that a withholding rule exists must remain distinct from the proposition that a specific payment falls within it. The existence of a low corporate tax rate must remain distinct from the operational question of whether the enterprise can place real functions, people, risks, and decision rights in that jurisdiction. By maintaining two universes, the architecture can connect law to facts through explicit claims rather than through unrecorded intuition.

The vault functions as the durable memory joining the universes. It combines human-readable Markdown with structured CSV and JSON records. Markdown makes the system inspectable by professionals; structured files make it testable by machines. The graph is portable, but portability does not imply informality. Stable identifiers, data dictionaries, hashes, decision records, validation outputs, and a current-state cache give the vault the properties of a governed analytical system.

At Step 0, the state is intentionally incomplete: there are zero tax-structure recommendations. That absence is a feature. It establishes a clean baseline from which the first reasoning experiment can be observed. Later steps never overwrite this state. Each stage copies the prior vault, adds new objects, validates the expanded state, and preserves the historical version.

The system's scale changes by Step 10. The current rule universe remains at exactly 100 modules, but version history contains 110 records after 10 modules change. Five new companies expand the enterprise universe to 15 conglomerates, 174 entities, 159 ownership edges, and 251 intercompany transactions. The architecture therefore demonstrates two different forms of growth:

- ▶ **diachronic growth**, in which an existing rule acquires a new version while the prior version remains preserved; and
- ▶ **synchronic growth**, in which new enterprises enter the current analytical universe.

The distinction matters because change should not be modeled as replacement alone. Institutional memory must retain the state that supported an earlier recommendation even after the current law or enterprise universe changes. Otherwise, the system cannot explain why a prior position was reasonable when made, determine which conclusions are stale, or distinguish a genuine revision from silent historical rewriting.

The experimental environment is therefore best understood as a temporal knowledge graph under governance. Nodes contain rules, entities, transactions, claims, recommendations, decisions, and permissions. Edges express ownership, flows, reliance, contradiction, support, limitation, review impact, and authority. Versions preserve time. Gates determine which transitions may become institutional state.

3. Conceptual Foundations

3.1 The exo-brain is a governed memory, not a tax oracle

A tax oracle answers a question such as "Where should the group place its intellectual property?" A tax exo-brain preserves the state required to evaluate that question responsibly. It remembers which rule versions were consulted, which business functions were assumed, which candidate hubs were considered, which contrary rules limited the analysis, which stress scenario exposed fragility, which evidence remained missing, which committee condition was imposed, and which actions remained prohibited.

This difference transforms the unit of intelligence. The unit is no longer the answer. It is the answer plus its lineage, uncertainty, decision status, and permission boundary.

3.2 The dual-universe ontology prevents category errors

Many analytical failures arise because legal propositions and enterprise facts are stored in the same undifferentiated narrative. The laboratory imposes an ontology in which the external universe and internal universe remain separate. A candidate tax structure becomes a bridge between them, not a fusion of them.

The bridge is constructed through claims. A claim may assert that a legal rule applies to a payment, that an entity performs a function, that a substance requirement can be satisfied, or that a projected economic result follows under specified assumptions. Because each claim has sources, dependencies, confidence, and possible contradictions, the system can identify precisely where law ends, facts begin, and interpretation connects them.

3.3 Optimization is constrained design, not scalar minimization

A candidate structure can improve one tax metric while becoming worse along several institutional dimensions. Lower cash tax may increase withholding leakage. A favorable effective tax rate may depend on fragile substance assumptions. A high five-year NPV may require operational complexity that the enterprise cannot govern. A low-rate hub may fail transfer-pricing, controlled-foreign-company, minimum-tax, anti-avoidance, documentation, or implementation constraints.

The laboratory therefore treats optimization as multi-objective and constrained. Economic measures are relevant, but they cannot compensate for a failed hard constraint. This is a general lesson for autonomous systems: some factors belong in a weighted score; others define the feasible set itself.

3.4 Provenance and contradiction are computational objects

Provenance is not a footnote added after reasoning. It is a set of immutable source snapshots, file hashes, effective dates, claim links, and recommendation dependencies. Contradiction is not noise to be averaged away. It is an object linking a supporting claim to a limiting claim, with severity, status, ownership, and consequences for reliance.

Once represented as objects, provenance and contradiction can affect computation. Confidence can be capped by source quality. A high-severity contradiction can place a proposal on hold. A

changed source can identify every dependent claim and recommendation. This is the point at which explainability becomes operational rather than rhetorical.

3.5 Recommendation, confidence, permission, approval, and implementation are distinct states

The architecture separates five concepts that are often collapsed.

- ▶ A **recommendation** is an analytical preference among alternatives.
- ▶ **Confidence** measures the strength and resilience of its support.
- ▶ **Permission** defines what analytical or operational use is currently allowed.
- ▶ **Approval** is an institutional decision accepting a bounded proposition.
- ▶ **Implementation** changes the external world.

A recommendation may exist at medium confidence. A committee may authorize additional diligence without approving the recommendation. A design may be selected for internal review without permission to transmit it. An adviser may be preferred without being appointed. An instruction pack may be complete while communication remains structurally impossible. These distinctions prevent analytical momentum from acquiring authority by accident.

3.6 Legal change must be represented as a versioned event

Continuous tax intelligence creates a danger: the system may treat the latest legal state as if it had always existed. The laboratory avoids this by versioning changed modules rather than overwriting them. The current universe remains 100 modules, while history expands to 110 records. Recommendations continue to point to the versions on which they were originally based.

This produces temporal accountability. The system can state both what is current and what was relied upon. It can identify affected subgraphs and create review queues without manufacturing Recommendation V2 automatically. Learning becomes accretion plus governed reconsideration, not historical erasure.

3.7 Governance is part of cognition

Governance is often described as a control layer surrounding an otherwise autonomous reasoning engine. In this laboratory, governance is internal to the reasoning architecture. A system that cannot distinguish "design selected" from "approved" does not understand the institutional meaning of its own output. A system that cannot prove that a dry-run message was not sent does not reliably distinguish thought from action. A system that silently overwrites Recommendation V1 when a rule changes is not continuously learning; it is continuously forgetting.

Institutional intelligence therefore requires both analytical competence and state-transition discipline.

4. The Eleven-Stage Maturity Ladder



Figure 1. The eleven Baby Steps form a maturity ladder in which capability and control advance together.

4.1 Baby Step 0: Build memory before judgment

Step 0 constructs the dual-universe vault. The external universe contains 20 jurisdictions and 100 tax-code modules. The internal universe contains 10 conglomerates, 154 entities, 144 ownership edges, and 226 intercompany transactions. Stable identifiers, portable Markdown, structured data, deterministic validation, and a current-state cache establish the baseline.

The decisive feature is the absence of recommendations. Decision DEC-000 authorizes reading, browsing, organizing, validating, and designing the Step 1 experiment. It expressly prohibits tax recommendations, real-jurisdiction inference, filings, transactions, restructuring, communications, and implementation.

This is not administrative caution. It creates epistemic cleanliness. If recommendations were embedded in the initial records, later reasoning could merely reproduce prior conclusions while appearing analytical. By setting the initial recommendation count to zero, the system can observe the moment at which interpretation first enters the state.

The current-state cache summarizes the active quarter, counts, latest decision, next permitted experiment, and explicit prohibitions. It reduces context reconstruction while preserving the full vault as durable memory.

Pedagogical lesson. The first capability of an institutional exo-brain is not inference but disciplined memory. The first governance act is to make the initial permission state narrower than the system's eventual ambition.

If this stage were skipped. Later recommendations would lack a stable universe, reproducible identifiers, temporal baseline, and defensible distinction between legal rules and enterprise facts.

4.2 Baby Step 1: Complete the first governed structure-design loop

Step 1 creates Recommendation V1 for each of the 10 conglomerates. Candidate hubs and design patterns are evaluated against the tax-code universe and the enterprise graph. The output includes regional operating-principal and treasury structures, substance-aligned intellectual-property and service hubs, distribution and procurement alignments, and regulated-services arrangements.

The portfolio scores range from 60.48 to 82.29, with medium or high confidence. Yet the scores are not treated as approvals. Exactly seven supporting or constraining rule links accompany each recommendation. Rate minimization is expressly subordinated to substance, anti-avoidance, documentation, and governance.

Each Recommendation V1 is a historical object. Step 0 remains unchanged, and every recommendation remains pending human review. The architecture therefore completes a governed loop:

observe -> normalize -> connect -> generate alternatives -> evaluate -> challenge -> recommend -> decide -> record -> remember

The recommendation register is valuable not because it identifies a final answer, but because it creates a versioned analytical baseline against which stress, evidence, and legal change can later be measured.

Pedagogical lesson. The first complete unit of institutional reasoning is a persistent loop, not a transient answer.

If this stage were skipped. The vault would be a sophisticated taxonomy without an inspectable decision-support process or a baseline recommendation version.

4.3 Baby Step 2: Test generalization, constraints, and fragility

Step 2 asks whether one design model remains defensible across heterogeneous business models and strategic objectives. Each Recommendation V1 is tested under eight controlled scenarios: two business-model tests, two strategic-objective tests, two binding-constraint tests, and two fragility shocks.

The experiment produces 80 scenario evaluations: 44 pass, 28 conditional, and 8 fail. Four recommendations are classified as robust, five as managed fragility, and one as fragile. Reduced substance capacity is the dominant portfolio-wide stress; an anti-avoidance and minimum-tax shock is the worst scenario for one financial-technology case.

Most important, the system refuses to let a high baseline score override a failed stress scenario. Delta Consumer Holdings, Evergreen Infrastructure, Frontier Energy Systems, and Ionis Media and Data qualify as robust. Five cases require mitigations. Cobalt Life Sciences is reopened before Step 3.

This stage introduces model-risk governance. It recognizes that a structure can look attractive under average assumptions while failing under a plausible change in operational capacity, legal constraints, or strategic priorities. The output is not Recommendation V2. It is a resilience classification and a disposition for the next analytical stage.

Pedagogical lesson. Generalization is demonstrated through heterogeneous stress, not repeated application of one scoring function.

If this stage were skipped. A portfolio could appear globally optimized while concealing business-model mismatch and substance fragility at the individual-company level.

4.4 Baby Step 3: Convert narrative tax reasoning into governed evidence

Step 3 freezes every relied-upon rule version, computes file hashes, creates recommendation-specific atomic claims, exposes unresolved tensions, and verifies complete paths from source to claim to recommendation to decision.

The resulting evidence architecture contains 44 immutable source snapshots, 70 atomic tax claims, 40 active claim dependencies, 40 unresolved contradictions, and 70 complete trace rows. The remaining 56 tax-code modules stay in the governed inventory but are not silently promoted into the analytical trace.

Every recommendation is fully traceable, but traceability does not imply reliability. Four recommendations are bounded and traceable, five are traceable with mitigations, and Cobalt is traceable but reopened. Each recommendation retains four high-severity unresolved tensions. Cobalt remains excluded from the Step 4 decision set pending redesign.

This distinction is fundamental. Provenance answers "Where did this proposition come from?" It does not answer "Is the proposition strong enough to act upon?" A weak claim can be perfectly traceable. A fragile structure can be fully documented. Traceability makes weakness visible; it does not cure it.

Pedagogical lesson. Explainability must expose the structure of reliance and allow contradiction to narrow permission.

If this stage were skipped. Committee products would compress assumptions, weak sources, and unresolved tensions into persuasive prose without a recoverable evidence chain.

4.5 Baby Step 4: Reconstruct governed state as a tax-committee product

Step 4 produces the first tax-committee memorandum, portfolio dashboard, exhibits, presentation, decision log, and data registers. The products are reconstructed from the governed state rather than independently authored from memory.

The committee is asked to authorize routing, not structures. Four robust recommendations advance to standard diligence. Five managed-fragility cases advance to enhanced diligence with mitigations and exception ownership. Cobalt remains on hold for substance-led redesign. Across the portfolio, 53 open committee conditions remain visible. No structure is approved and no real-world action is authorized.

The committee memorandum identifies operating substance as the dominant portfolio risk. It therefore requires Step 5 to test functions, people, premises, decision rights, and documentary capability before economic optimization. The presentation layer does not conceal the fact that all 10 recommendations retain unresolved tensions.

This stage shows that communication is itself a governance operation. A committee cannot consume hundreds of linked objects directly, but compression creates a risk of false certainty. Reconstructing the memorandum and exhibits from the same governed data reduces narrative drift.

Pedagogical lesson. A decision product should be a reproducible view of institutional state, not an elegant story assembled outside the system.

If this stage were skipped. The system could reason but would remain institutionally unusable, or human presenters would reintroduce omissions and unsupported confidence.

4.6 Baby Step 5: Make diligence a causal evidence-refresh process

Step 5 reconciles the 53 committee conditions against 159 new immutable synthetic evidence records. Each condition is supported by a document, an interview note, and an independent calculation reperformance. Records contain an owner role, as-of date, quality score, independence indicator, and content hash.

The results are intentionally differentiated. Four conditions close for each of the strongest cases; several managed-fragility cases close five of six conditions but retain one exception; Frontier and Ionis also retain exceptions. Cobalt closes none of its seven redesign conditions

and remains on hold. Confidence ranges from 48 for Cobalt to 89 for Evergreen.

Closure has a bounded meaning: the evidence standard for the internal model has been met. It does not approve the underlying structure. Exceptions retain owners and remain visible in Step 6. Redesign-required findings block optimization selection.

This transforms diligence from accumulation into causality:

committee condition -> evidence request -> immutable evidence -> refreshed claim -> contradiction disposition -> confidence update -> next permission

Pedagogical lesson. Diligence creates value when new evidence changes a governed dependency, not merely when more files are collected.

If this stage were skipped. Optimization would operate on unresolved facts, and documentary volume could be mistaken for evidentiary quality.

4.7 Baby Step 6: Optimize inside a feasible institutional set

Step 6 evaluates 30 alternative structures under 120 scenarios. Cash tax, effective tax rate, withholding leakage, five-year NPV, operating substance, legal robustness, documentation, and implementation complexity enter the comparison.

The stage does not ask for one scalar maximum. It distinguishes objectives from constraints. A low-rate design cannot dominate if it fails a legal or operating requirement. Each alternative is tested under a base case, coordinated rate increase, withholding shock, and substance-cost stress.

Nine conglomerates receive a preferred internal design artifact. Cobalt receives none because its redesign block remains active. Every selected artifact is labeled "design selected for internal review"; approval and implementation remain false.

This stage is the laboratory's clearest demonstration that optimization and authority are different. Computation can rank feasible candidates. It cannot decide that a candidate is institutionally accepted, communicable, or implementable.

Pedagogical lesson. Advanced optimization begins by defining what cannot be traded away.

If this stage were skipped. The system would possess evidence but lack a transparent bridge from evidence to economic and structural comparison. If constraints were treated only as score penalties, impermissible structures could win.

4.8 Baby Step 7: Design adviser markets with hard gates

Step 7 constructs a fictional universe of 32 advisory firms and screens them across 10 conglomerates, producing 320 conflict and independence records. Four workstreams are considered for each group: tax counsel, transfer pricing, valuation, and implementation assurance.

Candidates are ranked on technical capability, sector experience, data security, jurisdictional coverage, and fee discipline. Yet ranking occurs only after conflict and independence gates. A candidate cannot compensate for a disqualifying conflict through superior expertise or lower fees.

The resulting shortlist is a process recommendation, not an appointment. Even preferred firms remain subject to a fresh conflict check, verified recipients, negotiated scope, and human approval. Cobalt's advisers are limited to redesign scope.

The broader lesson extends beyond professional services. Autonomous systems often convert every consideration into a weight and then maximize a continuous score. Institutional processes contain prohibitions as well as preferences. Some deficiencies are tradable; others remove an option from the feasible set.

Pedagogical lesson. Selection requires both optimization and non-compensatory gates.

If this stage were skipped. The system could recommend a technically strong adviser whose conflict, independence, security, or mandate status makes engagement impermissible.

4.9 Baby Step 8: Prove the boundary between simulation and execution

Step 8 creates 40 disclosure-bounded instruction packs and 80 implementation-planning tasks. Every recipient is fictional and contains no real contact information. Tax counsel and transfer-pricing recipients receive controlled Tier 2 disclosure; valuation and implementation-assurance recipients receive minimum Tier 1 disclosure.

Each conglomerate receives an eight-stage plan covering governance, legal analysis, substance, valuation, documentation, systems, return-position preparation, and post-change monitoring. Five explicit control gates separate analytical preparation from implementation. The implementation gate is blocked for every case.

The implementation playbook requires mandate verification, recipient identity verification, refreshed conflict and independence checks, minimum-necessary disclosure, dependency dry runs, separate human approval at every gate, rollback planning, and post-change monitoring.

The central result is negative but crucial: zero messages sent, zero tasks executed, zero restructurings, zero filings, and zero external-system changes. The architecture does not merely contain the instruction "do not send." It omits the execution path.

Pedagogical lesson. Safe agentic design requires architectural non-action, not only polite prompting.

If this stage were skipped. A later communication or workflow interface could convert a research prototype into an action system without adequate identity, disclosure, conflict, and authorization controls.

4.10 Baby Step 9: Introduce continuous legal intelligence without automatic revisionism

Step 9 simulates a quarterly event. Exactly 10 of the 100 current tax-code modules change. Prior versions are preserved, so version history expands to 110 records while the current universe remains 100. Five new companies enter the system, expanding the portfolio to 15 conglomerates.

The system recomputes the impact graph rather than recalculating everything. Six existing recommendations are linked to changed rules and enter a high-priority human review queue. Five new companies enter a medium-priority intake queue. Historical recommendations are linked to new companies only as hypothesis-generating experience.

No Recommendation V2 is generated automatically. This restriction separates discovery from institutional revision. The system may identify that an accepted recommendation has become stale or vulnerable. It may gather affected claims and propose questions. It may not silently rewrite the historical baseline or manufacture a new institutional position.

The quarterly playbook freezes the prior quarter, captures new source snapshots, versions only changed modules, recomputes the impact graph, adds companies through controlled intake, uses experience only for hypotheses, and routes affected items to human review.

Pedagogical lesson. Continuous learning requires targeted recalculation, preserved history, and a governed revision gate.

If this stage were skipped. The exo-brain would become stale. If implemented carelessly, it would instead become revisionist, rewriting recommendations whenever the legal universe changed.

4.11 Baby Step 10: Make the system usable without making the interface authoritative

Step 10 integrates the cumulative architecture into a read-only operating application. It exposes 20 jurisdictions, 100 current tax-code modules, 110 historical code records, 15 conglomerates, 174 entities, 159 ownership edges, 251 intercompany transactions, 10 Recommendation V1 records, 10 quarterly code changes, and 11 human-review items.

The application is deliberately static. It reads a generated JSON snapshot and contains no server-side component, credentials, write endpoint, approval button, messaging channel, filing capability, or implementation action. Five roles differ only in reading and bounded export permissions.

Eight scheduler jobs describe the quarterly operating cadence, but each is a read-only preview requiring a human trigger. Twelve health checks validate counts, history preservation, review routing, read-only permissions, and the absence of implementation authority. The permission matrix denies change, approval, transmission, and implementation to every application role.

The interface therefore exposes governed state without becoming a second source of truth. The vault remains the system of record. Usability is added at the boundary, not allowed to overwrite the architecture beneath it.

Pedagogical lesson. The final maturity step is not autonomous action; it is disciplined usability under a verifiable permission contract.

If this stage were skipped. The architecture would remain a research artifact. If usability were added without read-only controls, the interface could become an ungoverned operating system.

5. A Formal Model of the Exo-Brain

Let the governed institutional state at time t be:

$$S_t = (L_t, E_t, K_t, R_t, C_t, D_t, P_t, H_t). \quad (1)$$

where L_t is the versioned legal-rule universe; E_t is the enterprise universe of groups, entities, ownership, functions, assets, risks, and transactions; K_t is the evidence and claim graph; R_t is the set of recommendation and design objects; C_t is the contradiction set; D_t is the decision register; P_t is the permission state; and H_t is immutable history.

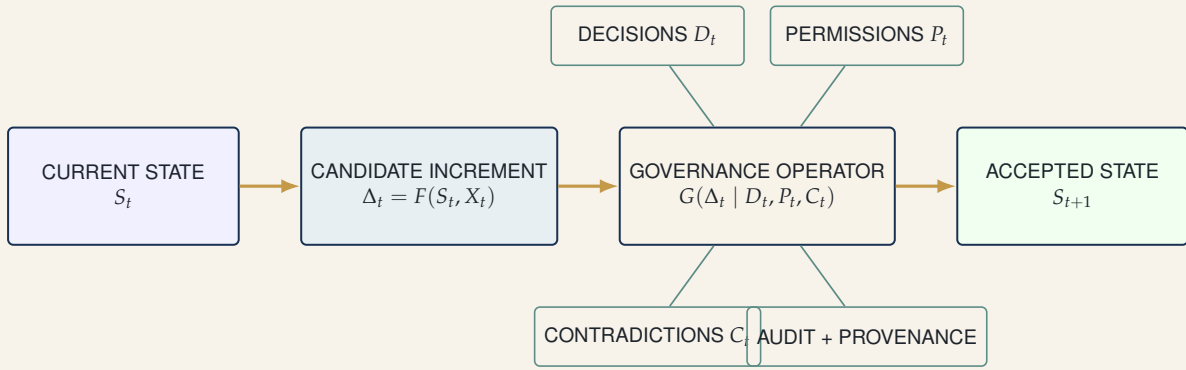


Figure 2. Analytical output becomes institutional state only after a governance operator applies decisions, permissions, contradictions, and provenance.

5.1 Candidate structure generation

For conglomerate i , the system generates a set of candidate structures:

$$A_i = F(L_t, E_{i,t}, O_i, B_i). \quad (2)$$

where O_i contains strategic objectives and B_i contains binding constraints. Candidate generation expands the possibility set; it does not confer feasibility or authority.

5.2 Multi-objective evaluation

For candidate $a \in A_i$, define an analytical utility:

$$U_i(a) = \sum_{k=1}^m w_{ik} u_{ik}(a) - \sum_{j=1}^n \lambda_{ij} q_{ij}(a). \quad (3)$$

The u_{ik} terms include cash tax, effective tax rate, withholding leakage, five-year NPV, substance quality, legal robustness, documentation quality, and manageable complexity. The q_{ij} terms are soft penalties. Weights w_{ik} and penalties λ_{ij} may vary by business model.

Hard constraints define the feasible set:

$$\mathcal{F}_i = \{a \in A_i : g_{ir}(a) \leq 0 \ \forall r, \ h_{is}(a) = 0 \ \forall s\}. \quad (4)$$

Only $a \in \mathcal{F}_i$ may be ranked. Conflict, independence, legal permissibility, minimum substance, mandatory documentation, and active redesign blocks belong in the feasible-set definition when failure cannot be compensated by other benefits.

5.3 Stress and resilience

Let Ω_i be the controlled scenario set for group i . The resilience of candidate a may be represented as:

$$\rho_i(a) = \min_{\omega \in \Omega_i} U_i(a \mid \omega). \quad (5)$$

A robust recommendation requires more than a favorable expected score. It must survive

the worst material scenario above a governed threshold and avoid a hard-constraint failure. The Step 2 classifications can therefore be expressed as a function of baseline value, stressed distribution, and constraint status:

$$\text{Class}_i = \Phi(U_i(a), \bar{U}_i(a), \rho_i(a), \text{violations}_i). \quad (6)$$

5.4 Evidence-bounded confidence

Let claim c depend on sources $Z(c)$. A simplified confidence function is:

$$\gamma(c) = \min \left(\tilde{\gamma}_{\text{type}(c)}, \sum_{z \in Z(c)} \alpha_z Q(z) - \beta I(c) - \delta X(c) \right). \quad (7)$$

where $Q(z)$ reflects authority, independence, recency, directness, and corroboration; $I(c)$ measures unresolved inferential distance; $X(c)$ measures contradiction severity; and $\tilde{\gamma}_{\text{type}(c)}$ caps confidence by claim type. An assumption cannot become equivalent to an independently corroborated fact merely through repeated use.

Recommendation confidence is bounded by its weakest material dependency:

$$\Gamma(R_i) \leq \min_{c \in \text{MaterialClaims}(R_i)} \gamma(c). \quad (8)$$

5.5 Governance as a state-transition operator

Analytical computation proposes a candidate increment:

$$\Delta_t = F(S_t, X_t). \quad (9)$$

where X_t may contain new evidence, scenarios, legal changes, committee instructions, or enterprise intake. The candidate increment becomes institutional state only through a governance operator:

$$S_{t+1} = G(\Delta_t \mid D_t, P_t, C_t, K_t, H_t). \quad (10)$$

The operator may accept, qualify, hold, reject, or route the increment for further work. It must preserve history, enforce active prohibitions, and record the decision that legitimizes the transition.

5.6 Targeted legal-change propagation

Let $G_t = (V_t, E_t)$ be the dependency graph. When a set of rule nodes ΔL_t changes, the affected analytical subgraph is:

$$\mathcal{A}_t = \{v \in V_t : \exists \ell \in \Delta L_t \text{ such that } \ell \rightsquigarrow v\}. \quad (11)$$

Only claims, recommendations, and cases reachable from the changed rules require immediate impact review. This targeted propagation reduces unnecessary recalculation while preserving

accountability.

5.7 Permission as a lattice

Permissions are not binary. They form an ordered set such as:

$$\text{READ} \preceq \text{ANALYZE} \preceq \text{PROPOSE} \preceq \text{DECIDE} \preceq \text{COMMUNICATE} \preceq \text{IMPLEMENT}. \quad (12)$$

Capability at level k does not imply permission at level $k+1$. Evidence may narrow permission automatically when confidence falls or contradictions rise. Widening permission requires an explicit authorized decision.

This asymmetry is one of the architecture's most important safety properties.

6. What the Laboratory Teaches About Autonomous Systems

6.1 Intelligence can grow through governed accretion

The system learns by adding objects and relationships to durable memory: a new source, a new claim, a resolved condition, a contradiction disposition, a changed rule version, a review event, or a reusable playbook. This differs from learning understood only as model-parameter adjustment. Institutional learning can occur through governed accretion of external memory.

The advantage is inspectability. One can observe exactly what changed, why it changed, and which conclusions it affects.

6.2 The hot cache solves continuity without replacing the vault

Long-running professional work cannot depend on reconstructing context from the beginning of every session. The current-state cache records counts, accepted decisions, open prohibitions, pending reviews, and the next permitted experiment. It behaves like a compact state vector for the next interaction.

The cache does not replace durable memory. It tells the system where it is; the vault preserves how it arrived there.

6.3 Human review is a transition mechanism, not ceremonial oversight

"Human in the loop" is often used vaguely. In this laboratory, the human gate has a precise function: it determines whether a candidate analytical increment may become accepted institutional state and whether permissions may widen.

The gate is meaningful only if the reviewer receives traceable evidence, alternatives, contradictions, stress results, and the exact proposed permission change. A button labeled "approve" is not governance if the reviewer cannot understand the state transition.

6.4 A safe system must represent both preferences and prohibitions

Weighted scoring is useful for choosing among acceptable options. It is dangerous when every institutional rule is converted into a compensatory weight. Conflict, independence, recipient identity, authority, and implementation permission are often hard gates.

The architecture therefore combines ranking with feasibility. This is a general model for autonomous procurement, finance, compliance, medicine, and law.

6.5 Negative capability is an engineering achievement

Step 8 demonstrates that a system can produce sophisticated instruction packs, implementation plans, and recipient matrices while remaining unable to transmit or execute them. Step 10 shows that an interface can expose the entire state while remaining unable to change it.

These are not missing features. They are proof that the architecture can separate deliberation capability from execution capability.

6.6 Continuous intelligence must resist revisionism

When rules change, the system identifies affected recommendations but does not automatically create new accepted positions. This preserves the difference between "the environment changed" and "the institution changed its view."

An autonomous system that silently rewrites its past cannot be audited. A governed system preserves both the historical conclusion and the current reason for reconsidering it.

6.7 The deepest product is institutional memory

The visible outputs are recommendations, committee memoranda, optimization reports, adviser shortlists, and dashboards. The deeper product is a memory architecture that reduces the loss of reasoning between quarters, teams, advisers, and decision cycles.

It preserves not only what the institution concluded, but why, under which assumptions, against which alternatives, with what unresolved tensions, and under what authority.

7. The Architecture of Authority

The laboratory separates four layers that are frequently collapsed in AI-enabled professional work.

1. **Information.** Tax-code modules, enterprise records, contracts, evidence, source snapshots, and historical versions.
2. **Interpretation.** Claims, dependency paths, scenarios, candidate structures, scores, confidence, and contradictions.
3. **Institutional position.** Accepted recommendations, committee routing decisions, diligence dispositions, preferred internal design artifacts, review queues, and permissions.
4. **External action.** Adviser engagement, disclosure, communication, restructuring, filings, transactions, and implementation.

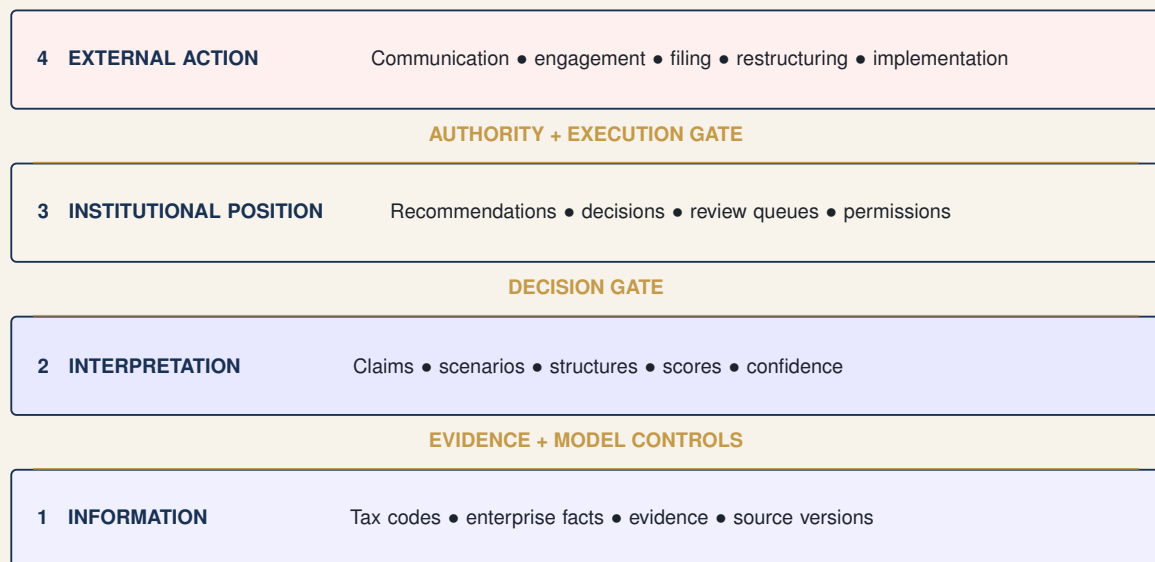


Figure 3. Tax information, interpretation, institutional position, and external action are separated by distinct governance gates.

Movement across layers requires different governance.

Information does not become interpretation merely because it has been retrieved. A rule and a company fact require a reasoned claim connecting them. Interpretation does not become institutional position because it has a high score. It requires evidence, challenge, and an authorized decision. Institutional position does not create external authority. Communication and implementation require identity, mandate, disclosure, conflict, security, approval, and execution controls.

The permission model should therefore be directional. Deteriorating evidence or a new contradiction may automatically narrow use. Widening use requires a human gate. This creates a conservative institutional bias: the system may become more cautious through computation, but it may not become more powerful through computation alone.

The Step 10 permission matrix makes the architecture executable. Every role can read some portion of the state; some may export bounded views. No role may change, approve, transmit, or implement. The interface does not merely describe policy. It denies write paths at the application boundary.

This architecture prevents three category errors.

- **Data is not advice.** A current rate or rule module does not determine a filing position.
- **Analysis is not approval.** A preferred design artifact is not an accepted institutional structure.
- **Approval is not execution.** Even an accepted internal position would require separate authority, recipients, controls, and operational steps before external action.

8. Why the Order of the Baby Steps Matters

The sequence is cumulative because each capability introduces a new failure mode that the next stage governs.

- ▶ Memory makes reasoning possible, but reasoning introduces hidden assumptions and favorable-selection bias.
- ▶ Versioned recommendations make comparison possible, but a common model may conceal business-model mismatch.
- ▶ Stress testing exposes fragility, but scenario results may remain persuasive abstractions without evidence lineage.
- ▶ Provenance makes reliance visible, but complex governed state must be compressed for institutional decision.
- ▶ Committee products enable routing, but committee conditions require disciplined evidence refresh.
- ▶ Diligence strengthens claims, but stronger evidence still needs transparent multi-objective design.
- ▶ Optimization creates preferred alternatives, but selection of advisers introduces conflict and independence risk.
- ▶ Provider shortlists make outreach conceivable, but recipient identity and disclosure controls must exist before communication is possible.
- ▶ Dry-run instruction proves non-action, but the external legal environment continues to change.
- ▶ Continuous legal intelligence identifies impact, but dynamic state needs a usable and testable application boundary.

The sequence follows a general design rule:

Design rule

Do not add a capability until the architecture can represent and govern the new failure modes that the capability introduces.

This is why the project repeatedly preserves Recommendation V1. Richer evidence, new scenarios, preferred designs, and changed rules may justify reconsideration. They do not themselves authorize revision. The restriction forces the system to distinguish analytical learning from institutional decision.

The order also prevents premature optimization. Economic comparison appears only after the system has separated universes, established recommendation lineage, tested fragility, frozen provenance, recorded contradictions, obtained committee routing, and refreshed evidence. This is not inefficient. It is the cost of ensuring that the objective function operates on governed inputs inside a feasible set.

9. Evaluation: What the Prototype Demonstrates

The completed laboratory demonstrates seven important properties.

9.1 Ontological separation

The rule universe and enterprise universe remain distinct. Legal modules, company facts, transactions, and functions are connected through explicit claims rather than conflated in prose.

9.2 Traceability

Recommendations can be traced to alternatives, enterprise facts, relied-upon rule versions, atomic claims, contradictions, scenarios, committee conditions, and decisions. The 44 frozen source snapshots and 70 trace rows show that provenance is selective and recommendation-specific.

9.3 Version integrity

Step 0 remains unchanged after Recommendation V1 is created. Prior rule versions remain preserved after the quarterly update. Recommendation V1 remains historical even when review becomes necessary. The architecture therefore supports temporal explanation.

9.4 Explainable adaptation

The system adapts across business models through group-specific objectives, constraints, stress tests, and evidence. It adapts over time through an impact graph and targeted review, not through unrecorded recalculation.

9.5 Permission-aware intelligence

Confidence, contradiction, committee decisions, redesign blocks, conflict gates, disclosure tiers, and application roles change permitted use. High analytical value never automatically widens authority.

9.6 Reproducible professional products

Registers, committee memoranda, exhibits, diligence reports, optimization outputs, selection reports, playbooks, and review queues are reconstructed from governed objects. Presentation remains connected to the state it summarizes.

9.7 Structural non-action

The system proves zero transmissions, zero appointments, zero filings, zero restructurings, and zero implementations. The read-only application contains no credentials, write endpoint, approval button, or messaging channel.

These achievements should be interpreted carefully. They demonstrate architectural behavior, not tax accuracy. The prototype is successful as a laboratory because it makes memory, lineage, uncertainty, and authority inspectable.

10. Limitations and Research Boundaries

The laboratory is an architecture experiment, not a production tax system.

First, all jurisdictions, rules, rates, entities, transactions, advisers, recipients, and outcomes are synthetic. The prototype does not demonstrate doctrinal accuracy, empirical predictive power, or defensibility before a real authority.

Second, the scoring, confidence, and scenario functions are pedagogical. A production system would require expert calibration, back-testing, sensitivity analysis, model-risk governance, and jurisdiction-specific legal review. Numeric transparency is valuable, but transparency does not validate the numbers.

Third, the five-module rule taxonomy is intentionally compressed. Real tax analysis would require treaties, domestic legislation, regulations, administrative guidance, case law, accounting standards, customs, indirect tax, payroll, permanent establishment, beneficial ownership, anti-hybrid rules, interest limitations, dispute resolution, reporting obligations, and interaction effects.

Fourth, real enterprise facts are contested, incomplete, privileged, confidential, and temporally unstable. Evidence ingestion would require source authentication, retention policies, legal privilege controls, access segregation, data-residency rules, and defensible deletion.

Fifth, a real implementation involves behavioral and organizational dynamics that a graph cannot fully encode. People may not relocate, decision rights may not change in practice, systems may not support required documentation, and local management may resist formal designs.

Sixth, read-only architecture reduces execution risk but does not eliminate interpretation risk. A user can over-rely on a displayed score or misunderstand a synthetic recommendation. Interfaces must communicate assumptions, freshness, uncertainty, and permission clearly.

Seventh, the human decision layer can fail. An immutable approval record does not prove that the approver understood the evidence, possessed the correct mandate, or exercised independent judgment. Governance requires competence, incentives, accountability, and escalation rules, not merely workflow.

The appropriate claim is therefore bounded: the laboratory provides a reference architecture for governing institutional tax cognition.

11. Productionization Roadmap

A responsible transition from synthetic laboratory to controlled institutional pilot would require at least eight workstreams.

11.1 Canonical ontology and data contracts

Define production schemas for jurisdictions, legal authorities, effective dates, entities, ownership, functions, assets, risks, transactions, claims, recommendations, decisions, permissions, advisers, recipients, and audit events. Establish stable identifiers, mandatory fields, controlled vocabularies, and migration rules.

11.2 Authoritative legal-source ingestion

Build authenticated pipelines for legislation, treaties, regulations, guidance, rulings, and case law. Preserve publisher, jurisdiction, effective date, retrieval date, version, language, amendments, and supersession relationships. Separate similarity from legal authority.

11.3 Enterprise-data and privilege controls

Integrate entity management, finance, transfer-pricing, contracts, HR, treasury, and document systems under purpose limitation and least privilege. Classify confidential and privileged material, apply data-residency requirements, and log every access.

11.4 Claim and evidence governance

Require source-backed atomic claims, confidence caps by claim type, explicit inferential distance, contradiction ownership, freshness states, and mandatory evidence standards for material dependencies. Preserve the distinction between observed fact, management representation, model output, assumption, and legal interpretation.

11.5 Model and optimization governance

Validate candidate-generation methods, weights, stress scenarios, thresholds, objective functions, and hard constraints. Maintain model inventories, benchmark cases, drift monitoring, challenger models, override records, and independent review. Preserve the distinction between computational optimality and legal reliability.

11.6 Decision and permission services

Implement identity-aware approvals, role-based authority, dual-control gates, immutable decision records, permission inheritance, revocation, and escalation. Each state transition should specify who decided, what changed, what evidence was considered, and which permissions widened or narrowed.

11.7 Secure interface and controlled automation

Begin with read-only views and bounded export. Add drafting only after source, recipient, disclosure, and approval controls are tested. Add any communication or external action through separate tools with explicit mandates, recipient resolution, content approval, audit logging, and reversible deployment.

11.8 Operational assurance

Establish health checks, disaster recovery, cyber controls, incident response, vendor oversight, legal-change service levels, periodic access review, and red-team exercises. Test not only whether the system can complete a workflow, but also whether it remains inactive under ambiguous or unauthorized conditions.

Productionization should therefore proceed capability by capability. The correct question is not "When can the system be connected to email or filing software?" It is "What new state transition would that connection create, what evidence and authority justify it, and how can

the institution prove that the transition is governed?"

12. Conclusion: The Objective Is Governed Adaptation

The Tax Planning Exo-Brain laboratory offers a disciplined answer to a problem that will become increasingly important for professional institutions: how can an organization enlarge machine intelligence without allowing machine authority to enlarge silently?

The answer is architectural.

The system begins by separating law from enterprise facts and building durable memory before judgment. It creates versioned recommendations but subjects them to business-model stress and substance shocks. It decomposes fluent reasoning into sources, claims, confidence, dependencies, and contradictions. It reconstructs committee products from governed state. It turns diligence into evidence refresh and optimization into constrained design. It screens advisers through hard conflict and independence gates. It simulates instruction while proving that nothing was sent. It incorporates legal change through version history and targeted review rather than automatic revision. Finally, it exposes the accumulated state through a read-only interface whose lack of execution authority is a deliberate feature.

The cumulative achievement is not a machine tax adviser. It is something more institutionally plausible: a governed cognitive infrastructure.

Such infrastructure changes the economics of professional memory. It reduces the loss of reasoning between quarters and teams. It preserves why a recommendation was made, what assumptions it required, which alternatives were rejected, what evidence later changed, and which human authority permitted the next step. It allows new legal information to identify affected conclusions without rewriting unaffected history. It directs attention toward material change and makes uncertainty visible at the point of decision.

The deepest lesson is that advanced systems should be evaluated by more than the sophistication of their outputs. They should also be evaluated by the quality of their memory, the explicitness of their ontology, the visibility of their contradictions, the integrity of their versions, the separability of their authority layers, and their ability to remain inactive.

An institutionally mature exo-brain can know more tomorrow than it knows today. It can discover that yesterday's position has become fragile. It can prepare the evidence and alternatives for reconsideration. It can tell the appropriate human authority exactly what changed and why.

What it cannot do is pretend that learning, recommending, deciding, communicating, and implementing are the same act.

That distinction is the foundation of governed autonomy.

Appendix A. The Eleven Baby Steps at a Glance

Step	Capability introduced	Principal governed objects	Central control
0	Dual-universe durable memory	100 tax-code modules, 10 conglomerates, entity and transaction graphs, DEC-000	Observe and organize only; zero recommendations
1	First governed design loop	Recommendation V1, candidate hubs, supporting and limiting rule links	Internal analytical baseline; no approval
2	Generalization and fragility testing	80 scenarios, resilience classes, dispositions	High baseline score cannot override failed stress
3	Provenance and contradiction control	44 source snapshots, 70 claims, 40 dependencies, 40 tensions	Traceability records weakness; it does not cure it
4	Tax-committee decision product	Memorandum, exhibits, 10 routing decisions, 53 conditions	Route diligence only; zero structures approved
5	Controlled diligence	159 evidence records, condition closures, exceptions, confidence refresh	Closure is analytical, not implementation approval
6	Constrained optimization	30 alternatives, 120 scenarios, nine preferred design artifacts	Hard legal and substance constraints define feasibility
7	Adviser-process design	32 firms, 320 conflict records, four workstreams per group	Conflict and independence are non-compensatory gates
8	Controlled instruction dry run	40 instruction packs, 80 tasks, disclosure tiers, gate logs	Structural inability to transmit or implement
9	Quarterly legal intelligence	10 changed codes, five new companies, 11 review items	Targeted review; no automatic Recommendation V2
10	Read-only operating boundary	Manifest, scheduler, health checks, roles, static application	No write, approval, communication, filing, or implementation path

Appendix B. Source Materials

This monograph synthesizes 36 stage-level Markdown sources recovered from the Tax Planning Exo-Brain packages. The source set includes:

1. Step 0 package README, system README, roadmap, DEC-000, success validation, and current-state cache.
2. Step 1 package README and Recommendation V1 register.
3. Step 2 package README and resilience register.

4. Step 3 package README, traceability register, provenance register, and contradiction register.
5. Step 4 package README, tax-committee memorandum, and committee exhibits.
6. Step 5 package README and controlled diligence report.
7. Step 6 package README and constrained optimization report.
8. Step 7 package README and adviser-selection report.
9. Step 8 package README, controlled-instruction report, and implementation playbook.
10. Step 9 package README, quarterly intelligence report, and change-control playbook.
11. Step 10 package README, operating-system report, application manual, and read-only application README.
12. Master roadmaps and consolidated package READMEs spanning the complete Steps 0-10 sequence.

All legal rules, rates, jurisdictions, enterprises, advisers, recipients, evidence, and outcomes described in the laboratory are synthetic and intended solely for architecture testing and pedagogy.